

UNIVERSAL ON-DEVICE DEVELOPER SCAFFOLDING

Agentic Architect

Spoken Ideas on iQOO Smartphone → AI-Primed Repository on Laptop

Turns fleeting developer voice prompts into deterministic, production-grade repositories with full AI agent context manifests, RFC 7807 contracts, and automated IDE orchestration.

1. Mobile Layer (iQOO)

SNAPDRAGON NPU INFERENCE

- Voice / Audio Prompt Ingestion
- Qwen 2.5-Coder / Phi-4-Mini (INT4)
- Qualcomm GenieX SDK (45 TOPS)
- 100% Offline • Zero Cloud Latency

2. Core Manifest Engine

CONTEXT MANIFEST PYRAMID

- Zod Validation & RFC 7807 Schema
- AGENTS.md, .cursorrules, CLAUDE.md
- Domain Skills: API, UI & Tests
- bundle.zip Stream / File Writer

3. Green Light Handoff

DESKTOP AUTO-BOOT & PRIMING

- iQOO Office Kit folder transfer
- bootstrap.sh / .ps1 auto-unpack
- Automated npm install hydration
- Boots Cursor & Primes Cline / Aider

The "Red Light" Scenario & The Disconnected Developer

How to capture sudden technical breakthroughs when away from your keyboard without cloud fragility.

The Status Quo: Friction & Cloud Fragility

■ Inspiration Lost in Transit

Developers get architecture ideas during commutes, red lights, or walking. By the time they reach a laptop, mental context is lost.

■ Cloud LLM Latency & Cost

External cloud APIs (OpenAI, Claude) require continuous 5G connection, suffer round-trip network lag, and cost per token.

■ Security & IP Leak Risk

Sending internal proprietary schemas and architectural requirements over public APIs introduces compliance risk.

■ Cold Repository Start

Manually setting up directory trees, TypeScript configs, ESLint rules, and agent instructions takes 30-45 minutes per project.

The Agentic Architect Solution: 100% Offline NPU

■ Hands-Free Spoken Ingestion

Speak natural-language requirements directly into your iQOO phone: "FastAPI backend, Next.js frontend, Tailwind UI".

■ Snapdragon Hexagon NPU Acceleration

Runs quantized Qwen 2.5-Coder / Phi-4-Mini in ~160ms on-device via Qualcomm GenieX SDK with 0 cloud calls.

■ Strict Constrained Grammar Decoding

Constrains model output directly into verified JSON adhering to strict Zod and RFC 7807 schemas.

■ Instant 'Green Light' Handoff

Arrive at desk: iQOO Office Kit automatically syncs bundle.zip. Laptop boots IDE primed with all rules.

Snapdragon NPU Acceleration & Qualcomm GenieX SDK

On-device INT4 quantized execution delivers desktop-class code reasoning with zero cloud egress.

45 TOPS

NPU Peak Compute

Qualcomm Hexagon NPU

168 ms

Inference Latency

Sub-second on-device turn

1.85 GB

Memory Footprint

INT4 Quantized weights

58.2 tok/s

Generation Throughput

Instant grammar output

Qualcomm GenieX SDK Architecture

HARDWARE-AWARE MODEL COMPILATION

- **Graph Quantization (INT4 / INT8)**
Weights compiled using Qualcomm AI Model Efficiency Toolkit (AIMET) preserving code semantics.
- **Hexagon Direct Tensor Offload**
Bypasses CPU/GPU thermal limits for sustained, energy-efficient generation without throttling.
- **Grammar-Guided Constrained Decoding**
Regex/EBNF constraints guarantee that model tokens strictly parse into ProjectConfig JSON.
- **Dual Model Fallback Strategy**
Primary: Qwen 2.5-Coder 7B INT4 (complex stacks); Secondary: Phi-4-Mini 3.8B (ultra-low power).

Zero-Hallucination JSON Constraints

SYNTACTIC PARSING ENFORCEMENT

- **Zod Schema Invariant**
Ensures runtime, architecture, install/dev/test commands, and packages match schema.
- **Polyglot Stack Detection**
Automatically identifies frameworks (FastAPI, Next.js, Go Gin, Spring Boot, Vue, Tailwind).
- **Safe Boundary Pre-Configuration**
Embeds zero-secret rules, lockfile integrity, and non-destructive Git constraints into JSON.
- **Offline Kotlin Zip Streamer**
Packages manifest files directly to /sdcard/iQOO_Share/bundle.zip with zero network activity.

Manifest Pyramid & The "Green Light" Handoff

A universal context layer ensuring all AI developer assistants adopt deterministic project rules.

The Multi-Agent Manifest Pyramid

AGENTS.md [Cline, Roo Code, Aider]

Root commands (npm test, dev), safety boundaries, git rules

.cursorrules [Cursor IDE Composer]

Tech stack, typing standards, UI rules, anti-patterns

CLAUDE.md [Claude Code CLI]

Session initialization, build commands, test workflows

GEMINI.md [Google Antigravity]

Domain rules, RFC 7807 contracts, modular guidelines

skills/api-contracts.md [All Agents]

RFC 7807 problem details, unified success envelope schema

skills/ui-component-system.md [All Agents]

Tailwind CSS design tokens, accessibility (a11y), clean UI

skills/testing-patterns.md [All Agents]

Vitest & pytest test conventions, isolation, mock standards

The "Green Light" Transfer Bridge

CHANNEL 1: iQOO OFFICE KIT

- Peer-to-peer Wi-Fi / Bluetooth fast sync
- Direct drop into ~/iQOO_Share/bundle.zip
- Zero cloud account or third-party server

CHANNEL 2: LOCAL EXPRESS STREAM

- POST /api/export HTTP pipeline
- Archiver v8 in-memory stream
- Instant curl fetch over local Wi-Fi

RELIABILITY & ENCRYPTION

- In-memory streaming (0 temp disk leaks)
- SHA-256 integrity check upon receipt
- RFC 7807 error envelopes for all routes
- 100% deterministic template outputs

Laptop Auto-Boot & Multi-Agent Priming

Developer sits down, runs bootstrap.sh, and begins coding immediately with fully primed agents.

Step 1: Unpack & Hydrate

AUTOMATED WORKSPACE SETUP

- **bootstrap.sh / .ps1**
One-command terminal execution.
- **Archive Ingestion**
Detects local iQOO_Share/ or pulls from Express endpoint.
- **Clean Extraction**
Unpacks into ./scaffolded-workspace with exact folder structure.
- **Dependency Hydration**
Executes npm install automatically to install packages.
- **Manifest Verification**
Verifies AGENTS.md, .cursorrules, CLAUDE.md, and skills/ are intact.

Step 2: IDE Auto-Boot

INSTANT DEVELOPER CONSOLE

- **Cursor / VS Code Launch**
Script automatically launches cursor ./scaffolded-workspace.
- **Zero Manual Setup**
No need to create folders or write starter configuration files.
- **ManifestViewer.tsx**
Interactive visual console for adjusting manifests or prompts.
- **RFC 7807 Live Validation**
API endpoint contracts and test suites ready out-of-the-box.
- **Full Monorepo Support**
Express, Next.js, FastAPI, or Kotlin pre-configured.

Step 3: Multi-Agent Priming

DETERMINISTIC COLLABORATION

- **Cursor Composer**
Instantly adopts .cursorrules for type-safe code completions.
- **Claude Code CLI**
Reads CLAUDE.md to understand build, lint, and test commands.
- **Cline & Roo Code**
Strictly constrained by AGENTS.md guardrails and boundaries.
- **Aider Terminal**
Picks up repo architecture map for precise patch creation.
- **Zero Cold-Start Lag**
Agents start generating production code from second one.